Name _____ Date _____ Class _____

Notes & Lab: Potential and Kinetic Energy

NOTES

1. **Potential Energy** is _____.
2. An object with potential energy _____
3. Potential energy depends on an object's _____ or _____

   Example:

BrainPop: Write down at least 1 thing you learn from the video.

4. **Kinetic Energy** is the _____.
5. It depends on the _____ and _____ of a moving object.

   Example:

BrainPop 2: Write down at least 1 thing you learn from the video.

CARD SORT

Potential Energy	Kinetic Energy
Pictures in this category:	Pictures in this category:
Reason these pictures are grouped together:	Reason these pictures are grouped together:

LAB

Question: How are Potential Energy and Kinetic Energy related?

Hypothesis: If the marble is moved up the ramp then it will push the cup (circle one)
Farther/
Not as Far because

_____
_____
_____

Materials: 1 marble, 1 ramp, 5 textbooks, 1 cup, 1 ruler, 1 lab set-up sheet

Methods: 1. Use the sheet provided to set up your lab equipment. Be sure to follow any directions your teacher gives you.
2. Place the marble at the first line on the ramp (Height A) and release the marble.
3. Draw a line where the back of the cup stops (the side of the cup NOT touching the ramp).
4. Measure the distance between where the back of the cup started and where it stopped. Record this distance under Trial 1 below.
5. Repeat steps 2-4 for trails 2 and 3 at height A.
6. Find the average distance traveled from height A by adding the distances from the three trials together and dividing by 3.
7. Repeat steps 2-6 with the marble at the second line on the ramp (Height B).
8. Repeat steps 2-6 with the marble at the end of the ramp (Height C).
9. Be sure you record all information in the data table below.

Height	**Cup Distance** Trial 1	**Cup Distance** Trial 2	**Cup Distance** Trial 3	**Cup Distance** Average
A (1st line)	_____ cm	_____ cm	_____ cm	_____ cm
B (2nd line)	_____ cm	_____ cm	_____ cm	_____ cm
C (end of ramp)	_____ cm	_____ cm	_____ cm	_____ cm

Conclusions:

1. When do you think the marble had the most potential/stored energy? Why do you think this?

2. At what height did the marble push the cup the farthest? Why do you think this was?

3. After completing this lab, how do you think potential energy and kinetic energy are related?

LAST NOTES

6. Energy can be _____ from _____ to _____.
7. If an object has _____ to start, then it will have _____ when it starts moving.